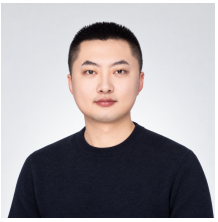
# Winston (Ze) Wang

Doctoral Researcher in Engineering

Kuala Lumpur, Malaysia

wangze_phd@163.com | +86 17602259373



Researcher in nonlinear dynamics and intelligent systems, with experience in conservative chaos, energy-based analysis, numerical modeling, and banking risk assessment.

## EDUCATION

**Taylor's University | Doctoral degree in Engineering, in progress**

2023-10 - Present | Expected graduation: 2027-10
High-dimensional conservative chaos; energy analysis; reinforcement learning; chaotic dynamics in LLM neural networks.

**Tiangong University | Master's, Control Science and Engineering**

2018-09 - 2021-03 | Conservative chaotic system modeling and energy analysis. GPA: 3.38.
Thesis: Analysis and circuit implementation of a conservative chaotic system based on a three-terminal memristor.

**Tiangong University | Bachelor's, Automation**

2014-09 - 2018-06 | Automation and control systems. GPA: 2.72.

## PROFESSIONAL EXPERIENCE

**Industrial Bank Co., Ltd. | 2021-12 - 2023-06**

Assessed individual and corporate customer risks; applied robotic process automation (RPA) to banking workflows; introduced satellite imagery and related technologies for post-loan monitoring of agricultural, photovoltaic, and wind-power projects.

## EARLY INDUSTRY EXPERIENCE

Huadian Inner Mongolia Energy, Tumed Power Generation Branch | 2018-10 - 2018-11
Operations and maintenance internship: plant monitoring, emergency repair participation, and PLC control-system maintenance.

Industrial site visits, Luoyang | 2013-05 - 2013-06
LYC, YTO, CRRC, and diesel-engine facilities; exposure to production lines, standardized operations, and information management.

## RESEARCH & PUBLICATIONS

Participated in the project on mechanical analysis, control, and applications of generalized rigid-body chaotic systems (Grant 61873186), and chaos/nonlinear-dynamics conferences in 2019 and 2020.

Wang, Z.; Qi, G. (2021). Modeling and Analysis of a Three-Terminal-Memristor-Based Conservative Chaotic System. Entropy 23(1), 71.

Hu, J.; Qi, G.; Wang, Z.; Chen, G. (2021). Rare Energy-Conservative Attractors on Global Invariant Hypersurfaces and Their Multistability. International Journal of Bifurcation and Chaos 31(3), 2130007.

Zhang, S.; Wang, Z.; Ma, D. (2021). Simulation and optimization of Flexsim shirt assembly line based on improved ant colony algorithm. Journal of Textile Research 42(3), 155-160.

Qi, G.; Hu, J.; Wang, Z. (2020). Modeling of a Hamiltonian conservative chaotic system and its mechanism routes from periodic to quasiperiodic, chaos and strong chaos. Applied Mathematical Modelling 78, 350-365.

## TOOLS & QUALIFICATIONS

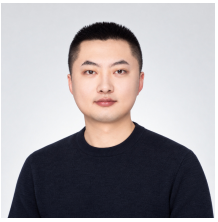
MATLAB, Altium Designer, Multisim, FlexSim, Microsoft Office; numerical simulation, mathematical modeling, circuit implementation, RPA, and satellite imagery.
National English Competition for College Students, Category C: Second Prize; CET-4; National Computer Rank Examination, Level 2; 2019-2020 academic scholarship (third tier).

## RESEARCH OUTLOOK

Investigating energy and chaos in large language models; exploring energy, momentum, and manifold geometry as potential routes to model optimization.

# 王泽

Winston (Ze) Wang
工程博士研究生 | 马来西亚吉隆坡
wangze_phd@163.com | +86 17602259373



从事非线性动力学与智能系统交叉研究，具有保守混沌、能量分析、数值建模与银行风险评估经验。

## 教育背景

### 泰莱大学 | 工程博士在读

2023-10 - 至今 | 预计毕业：2027-10
高维保守混沌建模与能量分析、强化学习，以及大语言模型神经网络中的混沌现象。

### 天津工业大学 | 控制科学与工程硕士

2018-09 - 2021-03 | 保守混沌系统建模与能量分析；GPA：3.38。
硕士论文：基于三端忆阻器的保守混沌系统分析与电路实现。

### 天津工业大学 | 自动化学士

2014-09 - 2018-06 | 自动化与控制系统；GPA：2.72。

## 工作经历

### 兴业银行 | 2021-12 - 2023-06

评估个人与对公客户风险，以机器人流程自动化（RPA）优化银行业务流程；引入卫星影像等技术，辅助农业、光伏和风电项目的贷后监测与管理。

## 早期行业经历

华电内蒙古能源有限公司土默特发电分公司 | 2018-10 - 2018-11
运行与检修实习：参与生产监控、应急抢修及PLC控制系统维护。

洛阳大型国企参访 | 2013-05 - 2013-06
观摩LYC轴承厂、一拖、中车及柴油机厂，学习流水线、标准化作业与信息管理。

## 学术研究与论文

参与“广义刚体混沌系统的力学分析、控制与应用”项目（批准号61873186）。
参加2019年与2020年混沌与非线性学术年会。

Wang, Z.; Qi, G. (2021). Modeling and Analysis of a Three-Terminal-Memristor-Based Conservative Chaotic System. Entropy 23(1), 71.

Hu, J.; Qi, G.; Wang, Z.; Chen, G. (2021). Rare Energy-Conservative Attractors on Global Invariant Hypersurfaces and Their Multistability. International Journal of Bifurcation and Chaos 31(3), 2130007.

Zhang, S.; Wang, Z.; Ma, D. (2021). Simulation and optimization of Flexsim shirt assembly line based on improved ant colony algorithm. Journal of Textile Research 42(3), 155-160.

Qi, G.; Hu, J.; Wang, Z. (2020). Modeling of a Hamiltonian conservative chaotic system and its mechanism routes from periodic to quasiperiodic, chaos and strong chaos. Applied Mathematical Modelling 78, 350-365.

## 工具与资格

MATLAB、Altium Designer、Multisim、FlexSim、Microsoft Office；数值仿真、数学建模、电路实现、RPA与卫星影像应用。
全国大学生英语竞赛C类二等奖；大学英语四级（CET-4）；全国计算机等级考试二级；2019-2020学业三等奖学金。

## 未来研究方向

深入研究大语言模型的能量与混沌特性，尝试从能量、动量与流形几何的角度探索模型优化的新路径。